Fiscal Year:	FY 2011	**Task Last Updated:**	FY 11/30/2011
PI Name:	Naidu, Mamta Ph.D.		
Project Title:	High LET-radiation induced DNA repair mechanisms in Oligodendrocyte progenitor cells (OPC) in vitro and in vivo		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mamta.naidu@griresearch.org	**Fax:**	FY
PI Organization Type:	NON-PROFIT	**Phone:**	(617) 444-3050
Organization Name:	GeneSys Research Institute/Tufts University School of Medicine		
PI Address 1:	Center for Cancer Systems Biology/Department of Medicine		
PI Address 2:	736 Cambridge Street, CBR 3		
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02135	**Congressional District:**	7
Comments:	NOTE: Previous affiliation was State University of New York-Stony Brook, from October 2011-December 2012. Prior to that, affiliation was Brookhaven National Laboratory, Upton, NY, until September 2011.		
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Space Radiobiology NNJ09ZSA001N
Start Date:	12/15/2009	**End Date:**	09/14/2011
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/14/2011 (from 12/14/2012) per PI's move to SUNY-Stony Brook in September 2011. Per A. Chu/ARC (Ed., 9/15/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA10DA32I		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA's radiation research program is focused on assuring that humans can travel in space without exceeding the limiting risks from radiation, and on advancing the understanding of response mechanisms to radiation-caused DNA damage. Radiation studies on the CNS have predominantly focused on neurons, with few studies directly addressing the role of glial cells. The focus of our research proposal will be to determine the major DNA repair pathway(s) induced by HZE radiation in glial cells. Oligodendrocytes (OL) are responsible for forming and maintaining myelin, the insulating sheaths of neuronal axons. The damage to OL causes neurological impairment as OL have a slow turnover and are repopulated by resident OL progenitor cells (OPC). Astronauts may encounter repeated exposure to charged particles over the course of their missions such as traveling to Mars and this exposure to low doses of cosmic rays has the potential to prevent astronauts from successfully meeting mission requirements. Base excision repair(BER) is one of the predominant DNA damage repair pathways up regulated in response to oxidative stress by low LET radiation. Also, Apurinic endonuclease 1(Ape1), a key BER player may be involved in adaptive response by linking repair to oxidative pathways, thus, it becomes important to study its role in fractionated versus single low dose of high LET radiation. As Ape1 is known to affect the G1/S transition in hematopoietic embryoid bodies (EB) cells, it also becomes important to determine if cell cycle dependent changes in Ape1 levels can be correlated to differentiation capacity of CG-4. Thus, our project will determine the effects of high LET radiation on the nuclear and mitochondrial BER in two systems; the central glia-4 (CG-4), a glial progenitor cell line that can differentiate into OL or astrocytes, and in OPC cells isolated from rat spinal cord. [Ed. note: Grant continues at SUNY-Stony Brook per PI's move in September 2011. See task, "High LET-radiation induced DNA repair mechanisms in Oligodendrocyte progenitor cells (OPC) in vitro and in vivo--NNX11AO89G."]
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our proposed study of the effects of fractionated, low dose versus single low dose of charged particles on glial cells of brain would help us understand the risk to astronauts exposed to chronic radiation exposures and the risk of normal tissues of patients receiving fractionated doses to their brain tumors. Our study will answer a critical question, "Would lower doses of higher Z particles produce more subtle latent changes, for example, changes in CNS cell population type that do not produce externally obvious deficits until months or years after exposure?" The oxidative stress produced by particles affects DNA repair capacity and our studies will show which DNA repair pathways are affected and how they affect the cells' responses to stress, so that we understand what risks are there for our astronauts on long term missions.
Task Progress:	ANNUAL REPORT--FY2011. Submitted August 2010. In the first year we have been able to show some good progress with specific Aims 1 and 2 when we show that fractionated low LET radiation caused a significantly higher induction of DNA repair enzyme Ape1 as compared to single dose, and preliminary studies indicate a correlation between levels of Ape1 induction and differentiation of rat glial cell line into higher astrocytes and immature oligodendrocyte, with lower mature oligodendrocyte formation. With regards to Aim3, where we proposed to study the same effects in vivo in rat spinal cord model, we are in process of low LET radiation exposures, so we can get this data to plan our high LET studies. This Aim has been delayed owing to one major reason of lack of personnel as my PDF hire started this September, despite me requesting for PDF last December! SUPPLEMENTAL INFORMATION FOR FINAL REPORT FOR NNA10DA32I—BROOKHAVEN NATIONAL LAB. Submitted November 2011. In the second year we have been able to show very good progress with specific Aims 1 and 2 when we show that fractionated high LET radiation causes a significantly higher induction of DNA repair enzyme Ape1 as compared to single dose, but this induction was significantly lower than that obtained with X-rays. Also Ape1 mitochondrial co-localization was more dose than fraction dependent with HZE. With Si fractionation, very few of the CG4 survived in differentiation media for less than a week. With proton fractionation, higher immature OL formation with some astrocyte formation, but again very few of these cells survived past 2-3 weeks in differentiation media. Almost no mature oligodendrocytes were detected with either Si or Proton exposures. Oxidative stress measurements revealed corresponding increase with fractionation with X-rays and protons and while the increase with HZE was more than 10 fold, the increase with HZE fractionation seemed variable. With regards to Aim3, where we proposed to study the same effects in vivo in rat spinal cord model, we obtained data on low LET radiation exposures, with 5 Gy of X-rays showing possible demyelination by 1.5 month after exposure and with 250 MeV protons, after 3 months of exposure, we got higher proliferation of ependymal cells of central canal indicating spinal cord injury over time. We are awaiting third repeat of Si and Fe ions from NSRL 11C for CG4 cells and second proton/ first HZE ion exposure for spinal cord to submit our two papers from this grant.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
Abstracts for Journals and Proceedings	Agarwal R, Naidu MD. "Crystal structure of Human Apurinic/Apyridinimic Endonuclease, Ape1 in a new crystal form." Protein Data Base. An Information Portal to Biological Macromolecular Structures. Protein Data Base. An Information Portal to Biological Macromolecular Structures, October 2011. http://www.pdb.org/pdb/explore/explore.do?structureId=3U8U , Oct-2011
Articles in Peer-reviewed Journals	Naidu MD, Mason JM, Pica R, Fung H, Pena L. "Radiation resistance in glioma cells determined by DNA damage repair activity of Ape1/Ref-1." Japanese Journal of Radiation Research 2010;51(4):393-404. http://dx.doi.org/10.1269/jrr.09077 , May-2010
Articles in Peer-reviewed Journals	Naidu MD, Agarwal R, Pena LA, Cunha L, Mezei M, Shen M, Wilson DM 3rd, Liu Y, Sanchez Z, Chaudhary P, Wilson SH, Waring MJ. "Lucanthone and its derivative Hycanthone inhibit Apurinic endonuclease-1 (APE1) by direct protein binding." PLoS One. 2011;6(9):e23679. Epub 2011 Sep 15. PubMed PMID: 21935361 , Sep-2011